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Mapping a Carrington Storm

Jeffrey J. Love¹ , Greg M. Lucas² , Anna Kelbert³ , E. Joshua Rigler¹ ,
Paul A. Bedrosian⁴ , and Neesha R. Schnepf^{1,2} 

Key Points:

- A Carrington-class storm would likely induce high-strength geoelectric fields in the United States East and Midwest
- More than 5.00 V/km would likely be induced in many places; in Virginia, strengths would range from 30.30 V/km to 0.05 V/km
- A Carrington-class storm would likely induce geoelectric-field strengths 55% greater than those of the 13–14 March 1989 storm

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

J. J. Love,
jjlove@usgs.gov

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Author Contributions:

Conceptualization: Jeffrey J. Love, Greg M. Lucas, Anna Kelbert, E. Joshua Rigler, Paul A. Bedrosian, Neesha R. Schnepf
Data curation: Jeffrey J. Love
Formal analysis: Jeffrey J. Love
Investigation: Jeffrey J. Love
Methodology: Jeffrey J. Love, Greg M. Lucas, Anna Kelbert, E. Joshua Rigler, Paul A. Bedrosian, Neesha R. Schnepf
Project administration: Jeffrey J. Love
Resources: Jeffrey J. Love
Software: Jeffrey J. Love, Greg M. Lucas
Supervision: Jeffrey J. Love
Validation: Jeffrey J. Love

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¹U.S. Geological Survey, Geologic Hazards Science Center, Denver, CO, USA, ²Laboratory for Atmospheric and Space Physics, University of Colorado, Boulder, CO, USA, ³Harvard and Smithsonian Center for Astrophysics, Cambridge, MA, USA, ⁴U.S. Geological Survey, Geology Geophysics and Geochemistry Science Center, Denver, CO, USA

Abstract A map is presented of median 1-min-resolution peak geoelectric-field strength across the United States as would be induced by magnetic storms as intense as the 2 September 1859 Carrington storm. The map is constructed from two data sets: Magnetometer time series from 22 ground-based observatories recording 40 magnetic storms, and surface impedance tensors derived from magnetotelluric measurements acquired at 1616 survey sites across the contiguous United States. Carrington-class storm geoelectric fields are likely to be very strong in the United States East and Midwest; >5.00 V/km at many places. In Virginia, strengths would likely range from 30.30 V/km, with a 68% confidence interval of [19.44,47.20] V/km, to as low as 0.05 [0.03,0.07] V/km. Comparison of model geopotentials with those measured on 30 long lines, indicates errors of about 18%. A Carrington-class storm would likely induce geoelectric fields with strengths 55% greater than for the 13–14 March 1989 storm.

Plain Language Summary The 2 September 1859 Carrington storm was one of the most intense magnetic storms ever recorded by magnetometers. It disrupted telegraph communication around the world. Should a storm similar to the Carrington storm occur today, geoelectric fields induced in the Earth could disrupt electric-power transmission, damage power-grid systems, and cause blackouts that would carry significant economic costs. To inform projects for improving the resilience of electric-power-transmission systems, a map is presented of the strengths of geoelectric fields such as would be induced across the United States by a Carrington-class magnetic storm. Geoelectric fields are likely to be strongest in the East, in the vicinity of many large cities, and in the Midwest. Geoelectric-field strengths would exceed, in many places, benchmark values used in testing the resilience of power-transmission systems.

1. Introduction

The magnetic storm of 2 September 1859 was one of the greatest ever recorded by magnetometers. Often called the “Carrington storm,” in honor of Richard Carrington, the astronomer who speculated that a solar flare he saw the day before might have been related to the storm’s cause (e.g., Clark, 2007; Cliver & Keer, 2012), it brought aurorae to many nighttime skies (e.g., Green et al., 2006; Hayakawa et al., 2019; Silverman, 2006), and it disrupted telegraph communication around the world (e.g., Boteler, 2006). Today, the Carrington storm is often regarded as the archetype of an extreme space-weather event (e.g., Cliver & Dietrich, 2013; Riley et al., 2018; Usoskin et al., 2023). The future occurrence of a storm of similar intensity could adversely affect many types of technological systems (e.g., Daglis, 2005; Hapgood et al., 2020) and carry substantial economic costs (e.g., Baker et al., 2008; Eastwood et al., 2017; Oughton et al., 2019). This expectation has been reinforced through the cumulative experience garnered from subsequent magnetic superstorms. The storm of 13–16 May 1921, e.g., rivaled the Carrington storm in intensity (e.g., Kappenman, 2006; Love et al., 2019b), and it disrupted telegraph and telephone communication around the world (e.g., Hapgood, 2019; Love, Lucas, Kelbert, Schnepf, et al., 2025). More recently, the 13–14 March 1989 storm wreaked havoc on radio communication, satellite operations, positioning systems, and geophysical surveys (e.g., Allen et al., 1989; Cliffswallow et al., 1993). It also caused the collapse of the Canadian Hydro-Québec electric-power system (Bolduc, 2002; Boteler, 2019; Guillon et al., 2016), interfered with power transmission in the Eastern United States (e.g., Kappenman, 2010; Love et al., 2022), and damaged a high-voltage transformer at a nuclear-power plant in New Jersey (e.g., Subudhi et al., 1994).

The physical connection between magnetic storms and interference experienced on long-line communication and power-transmission systems is qualitatively understood (e.g., Abda et al., 2020; Boteler, 2003; Piccinelli &

Visualization: Jeffrey J. Love,
E. Joshua Rigler
Writing – original draft: Jeffrey J. Love
Writing – review & editing: Jeffrey
J. Love, Greg M. Lucas, Anna Kelbert,
E. Joshua Rigler, Paul A. Bedrosian,
Neesha R. Schnepf

Table 1
Storms and Their Corresponding Intensities D

Cycle 22				Cycle 23				Cycle 24				Cycle 25			
Year	Mn	Dy	D (nT)	Year	Mn	Dy	D (nT)	Year	Mn	Dy	D (nT)	Year	Mn	Dy	D (nT)
1989	03	14	594	2003	11	20	427	2015	03	17	232	2024	05	11	415
1991	11	09	379	2003	10	30	384	2015	06	23	196	2024	10	11	340
1991	03	25	296	2001	03	31	382	2018	08	26	169	2025	01	01	208
1990	04	10	291	2004	11	08	372	2015	12	20	164	2024	08	12	204
1992	05	10	284	2000	07	16	301	2012	03	09	142	2023	04	24	198
1991	10	29	280	2000	04	06	297	2012	07	15	141	2023	11	05	158
1989	10	21	269	2001	11	06	283	2011	10	25	141	2024	10	08	157
1989	11	17	264	2001	04	11	265	2013	03	17	131	2023	03	24	155
1989	09	19	260	2005	05	15	245	2017	05	28	122	2024	09	17	124
1991	06	05	222	2000	08	12	235	2013	06	01	122	2024	09	12	124

Krausmann, 2014; Pilipenko, 2021). Briefly, geomagnetic field variation induces geoelectric fields in the Earth's electrically conducting interior. Storm-generated potential differences across the Earth's surface can, through ground connections, drive uncontrolled geomagnetically induced currents (GICs) in long-line systems. Two factors affect the geoelectric fields and corresponding surface potentials. First, electric currents in the ionosphere and magnetosphere, driven by the dynamic action of the solar wind, produce ground-level geomagnetic variation that can be both temporally and geographically complicated (e.g., Ngwira et al., 2015; Pulkkinen et al., 2006; Schillings et al., 2022). And second, because the conductivity structure of the Earth is complicated, surface electromagnetic impedance is complicated in geography and frequency dependence. Together, these factors result in storm-time geoelectric-field variation that can be highly time-dependent in strength and polarization, and it can have a high degree of geographic granularity (e.g., Bedrosian & Love, 2015; Cordell et al., 2021; McKay & Whaler, 2006; Murphy et al., 2021; Simpson & Bahr, 2020). As a result, predicting storm-induced geoelectric fields is a challenging goal for space-weather modeling projects (e.g., Pulkkinen et al., 2017; Thomson et al., 2010).

In this letter, we present a map of the median 1-min-resolution strength of the geoelectric field across the contiguous United States as would be induced by an ensemble of magnetic storms as intense as the 2 September 1859 Carrington storm. We construct the map by applying time-series and statistical analysis methods to two different data sets: Magnetometer time series from ground-based observatories recording magnetic storms that occurred over the past four solar cycles and surface impedance tensors derived from magnetotelluric measurements acquired at numerous survey sites across the United States. We validate the map against an independent third data set: Geopotential measurements made on long grounded lines during several intense magnetic storms. Our map informs projects for improving the resilience of electric-power-transmission systems against intense magnetic storms that might occur in the future.

2. Data

We use digital 1-min-resolution magnetometer data time series that were acquired at observatories operated by the United States Geological Survey (USGS) (e.g., Love & Finn, 2011) and Natural Resources Canada (e.g., Newitt & Coles, 2007). We select several-day intervals recording the 10 most intense magnetic storms, defined in terms of 1-hr resolution $D = \max\{-Dst\}$ (Karinen & Mursula, 2005; Mursula et al., 2008), during each of the four solar cycles 22–25, giving a total of 40 storms; see our Table 1. Our list encompasses storms with a range of intensities, including the 13–14 March 1989 storm (cycle 22, $D = 594$ nT), the 29–31 October 2003 storm (cycle 23, $D = 384$ nT) (e.g., Balch et al., 2004; Gopalswamy et al., 2005), and the 10–11 May 2024 storm (cycle 25, $D = 414$ nT). The least intense storms on our list are those of 1 June 2013 and 28 May 2017 (cycle 24, $D = 122$ nT).

We use long-period (10–10,000 s) USMTArray magnetotelluric transfer functions (electromagnetic impedance divided by the permeability of free space) derived from geomagnetic and geoelectric time series acquired simultaneously at 1616 survey sites during surveys that covered the contiguous United States with a nominal site spacing of 70 km (Schultz, 2010). The surveys were funded by the National Science Foundation EarthScope project (2006–2018) (Schultz et al., 2006–2018; Williams et al., 2010), National Aeronautics and Space Administration (2019) (Schultz et al., 2019, 2020), and the USGS (2020–2024) (Schultz et al., 2020–2024). We select high-quality transfer functions (minimal static offset) and such as to obtain nearly uniform geographic spacing. Generally, transfer-function amplitude is high over electrically resistive domains, such as igneous and metamorphic crust of the Appalachian Mountains in the Eastern United States, and the Superior Craton in the Upper Midwest. Transfer-function amplitude is generally low over regions with thick sedimentary cover, such as the Western Interior Seaway of the Great Plains and thick sedimentary basins along the Gulf Coast.

As parts of projects aimed at understanding magnetic-storm-induced interference, in the early 20th century, United States telegraph and telephone systems occasionally monitored geopotentials (voltages) between ground points on some of their long transmission lines. We use 30 peak geopotentials measured on long lines during six intense magnetic storms, a subset of a compilation given by Love and Wilbur (2023); see our Table 2. The most important of these measurements are 19 made during the 13–16 May 1921 storm ($D = 907$ nT) (Love et al., 2019b); these data are mostly from one source (American Railway Association, 1925). We excluded geopotentials measured on lines less than 100 km in length because we cannot resolve geopotentials for such short lines given the site spacing of the magnetotelluric surveys. We excluded measured geopotentials from Canada and near the United States–Canadian border because of the lack of regularly-spaced magnetotelluric-survey data in Canada.

3. Geoelectric and Geopotential Calculations

We obtain geoelectric-field estimates needed to construct our map of peak geoelectric-field strength for a Carrington-class magnetic storm across the United States using an extension of the methods documented in Lucas et al. (2020) and Love et al. (2022) and the data summarized in Section 2. Briefly: For each chosen magnetic storm, we interpolate the observatory magnetometer time series (e.g., Rigler et al., 2019; Vanhamäki & Juusola, 2020) to create 1-min time series of horizontal-component geomagnetic variation at each survey site. We Fourier transform the geomagnetic series to the frequency domain; multiply with the site's frequency-dependent magnetotelluric transfer function; and then inverse Fourier transform to obtain estimates of 1-min-resolution horizontal-component geoelectric-field time series for each survey site (Kelbert, 2020; Kelbert et al., 2017). For each survey site and for each storm with its intensity D , we identify and keep for analysis the estimated peak horizontal geoelectric-field strength $E_h(D)$ (which may not occur at the time of $D = \max\{-Ds\}$). Furthermore, for each storm, we obtain estimates of 1-min-resolution geopotential time series on each long line by integrating projections of the horizontal-component geoelectric-vector time series along each line's length. For each line and for each storm, we identify and keep for analysis the estimated peak geopotential $V(D)$ (which may not occur at the time of D , or the time of $E_h(D)$ for an individual site).

4. Geoelectric Strength Versus Storm Intensity

In Figure 1, we show peak 1-min-resolution estimated geoelectric-field strengths $E_h(D)$ for four example survey sites and for the intensities D of all 40 storms in Table 1. This plot shows generally increasing E_h for increasing D . Some sites have relatively high impedance and, therefore, strong geoelectric fields (VAQ58 (37.37°N, –77.58°E) in Virginia, MNC37 (47.82°N, –91.98°E) in Minnesota), whereas other sites have lower impedance and weaker geoelectric fields (KYQ51 (38.17°N, –83.02°E) in Kentucky, TXF28 (29.58°N, –103.15°E) in Texas). Here, we recognize that, in examining data that are positive and which cover a wide range of values, it can be useful to take logarithms of the data (e.g., Behrens, 1997; Morgenthaler, 2009; Tukey, 1977). We take the logarithmic transformation of the storm intensities and, for each site, the geoelectric strengths, $E_h(D) \rightarrow \ln E_h(\ln D)$. We fit straight-line models to these transformed values: $\ln \tilde{E}_h(\ln D) = a + b \cdot \ln D$, where a and b are constants for each site. The model fits, shown for the four example survey sites in Figure 1, illustrate that the trend of geoelectric strength versus storm intensity differs from one survey site to another. This reflects differences in site geomagnetic latitude and local frequency-dependent impedance. The dispersion of our estimated geoelectric-field strengths about our model indicates that, for an ensemble of storms, each with an identical intensity D , a range of $E_h(D)$ can be induced because of storm-to-storm differences in mesoscale geomagnetic variation.

Table 2
Peak Geopotential Measurements V and Model Peak Values $\hat{V}$ for Six Intense Storms

	Year	Mn	Dy	D (nT)	Source	Lat (°N)	Long (°E)	Lat (°N)	Long (°E)	V (V)	Source	$\hat{V}$ (V)
1903	10	31	513	Hayakawa et al. (2020)	Chicago, Illinois	41.88	-87.63	Minneapolis, Minnesota	-93.27	475	Western Electrician (1903b)	481
					Meadville, Pennsylvania	41.65	-80.15	Chicago, Illinois	-87.63	475	Western Electrician (1903b)	305
					St Paul, Minnesota	44.94	-93.09	Dickinson, North Dakota	-102.79	>300	Western Electrician (1903a)	695
					St Paul, Minnesota	44.94	-93.09	Dickinson, North Dakota	-102.79	960	American Railway Association (1925, p. 608)	497
1907	02	09	340	Love (2021)	Chicago, Illinois	41.88	-87.63	Minneapolis, Minnesota	-93.27	>200	Telephony (1909)	554
1909	09	25	595	Love et al. (2019c)	New York, New York	42.89	-78.88	Buffalo, New York	-74.01	80	Rhoads (1924, Figure 38)	194
1919	08	11	300	Love (2021)	Mobridge, South Dakota	45.54	-100.43	Miles City, Montana	-105.84	450	Railway Signal Engineer (1921)	153
1921	05	15	907	Love et al. (2019b)	Syracuse, New York	43.05	-76.14	Buffalo, New York	-78.88	450	American Railway Association (1925, p. 592)	146
					Minneapolis, Minnesota	44.98	-93.27	Fargo, North Dakota	-96.79	400	American Railway Association (1925, p. 592)	748
					Boise, Idaho	43.62	-116.20	Spokane, Washington	-117.42	400	American Railway Association (1925, p. 592)	163
					New York	42.65	-73.76	New York	-74.01	350	Rhoads (1924, p. 258)	876
					Rawlins, Wyoming	41.79	-107.23	Cheyenne, Wyoming	-104.82	300	American Railway Association (1925, p. 592)	55
					Jacksonville, Illinois	39.73	-90.23	Kansas City, Missouri	-94.58	220	American Railway Association (1925, p. 592)	523
					Harrisburg, Pennsylvania	40.27	-76.88	New York	-74.01	200	American Railway Association (1925, p. 592)	1158
					Pittsburg, Pennsylvania	40.44	-79.98	Columbus, Ohio	-83.00	200	American Railway Association (1925, p. 592)	261
					Big Spring, Texas	32.24	-101.48	El Paso, Texas	-106.49	200	American Railway Association (1925, p. 592)	140
					Dallas, Texas	32.78	-96.81	Lubbock, Texas	-101.89	200	American Railway Association (1925, p. 592)	158
					Creston, Iowa	41.06	-94.36	Omaha, Nebraska	-95.94	140	American Railway Association (1925, p. 592)	102
					Yuma, Arizona	32.69	-114.62	Los Angeles, California	-118.25	140	American Railway Association (1925, p. 592)	156
					Nashville, Tennessee	36.16	-86.77	Memphis, Tennessee	-89.97	140	American Railway Association (1925, p. 592)	228
					Nashville, Tennessee	36.16	-86.77	St. Louis, Missouri	-90.19	140	American Railway Association (1925, p. 592)	165
					El Paso, Texas	31.76	-106.49	San Antonio, Texas	-98.49	100	American Railway Association (1925, p. 592)	141
					Denver, Colorado	39.74	-104.98	Cheyenne, Wyoming	-104.82	45	American Railway Association (1925, p. 592)	77
1940	03	24	356	Oulu Dxt	San Antonio, Texas	29.42	-98.49	Dallas, Texas	-96.81	40	American Railway Association (1925, p. 592)	28
					Salt Lake City, Utah	40.76	-111.89	Butte, Montana	-112.52	15	American Railway Association (1925, p. 592)	181
					New York	42.35	-71.05	Binghamton, New York	-72.28	>800	J. A. Fleming (1940)	245
					Boston, Massachusetts	40.71	-74.01	Keene, New Hampshire	-75.91	800	Ireland (1940)	214
					St Louis, Missouri	38.62	-90.19	Springfield, Missouri	-93.29	>750	New York Times (1940)	165
					Minneapolis, Minnesota	44.98	-93.26	Fargo, North Dakota	-96.78	>400	Ireland (1940)	355
					Washington	38.91	-77.01	Cumberland WV	-80.60	400	Smith (1940)	273

A Shapiro and Wilk (1965) algorithm provides the probability (p -value) that the residual differences $\ln E_h(\ln D) - \ln \tilde{E}_h(\ln D)$ could be random realizations from a normal distribution. 98% of the 1616 survey sites have p -values greater than 0.05. We cannot, therefore, confidently reject the hypothesis that the (log-transformed) residuals for all sites are normally distributed. Going forward, then, we assume that, for each survey site, our model geoelectric-field strength $\ln \tilde{E}_h(\ln D)$ represents the median field strength of storms with intensity D . Additionally, we interpret the standard deviation σ of the residuals $\ln E_h(\ln D) - \ln \tilde{E}_h(\ln D)$ in terms of normal statistics. So, for example, given an ensemble of storms, each with an identical intensity D , 68% of geoelectric strengths will fall within the interval $[\ln \tilde{E}_h(\ln D) - \sigma, \ln \tilde{E}_h(\ln D) + \sigma]$. With exponential transformation, this interval is $[\tilde{E}_h(D)/\exp(\sigma), \tilde{E}_h(D) \cdot \exp(\sigma)]$, which has physical units (such as V/km). Confidence intervals have width factors $\exp(\sigma)$ of 1.55 for VAQ58, almost exactly the median factor across all sites. As an example, we estimate that the strengths of the geoelectric field at VAQ58 attained 18.83 V/km during the 13–14 March 1989 storm, whereas our model indicates the median field strength of storms with intensity $D = 594$ nT would be somewhat lower, 14.00 V/km. Our 68% confidence interval [8.99, 21.82] V/km encompasses the field strength we estimate was actually attained during that storm.

5. Carrington-Class Field-Strength Map

Geomagnetic field variation during the 2 September 1859 Carrington storm was measured at only one low-latitude observatory, that in Colaba, India. Using the magnetometer measurements (Fergusson, 1860; Hayakawa et al., 2022) made at Colaba, Love et al. (2024) developed a method for estimating the (1-hr-resolution) intensity for the Carrington storm, $D = 964$ nT. For each survey site, we plug this storm intensity into the models $\ln \tilde{E}_h(\ln D)$ to obtain extrapolated median geoelectric-field strengths for a Carrington-class storm. This extrapolation omits the unknown spatiotemporal complexity of the actual Carrington storm. The geoelectric strength is, here, the median of an ensemble of many hypothetical Carrington-class storms. We show a map of $\tilde{E}_h(D)$ in Figure 2, and in the Supporting Information S1, we show maps of $\tilde{E}_h(D)/\exp(\sigma)$ and $\tilde{E}_h(D) \cdot \exp(\sigma)$. Carrington-class strengths are greater than 10.00 V/km at 19 of the 1616 survey sites, and greater than 5.00 V/km at 68 sites. For comparison, 1.00 V/km is sometimes designated “extreme” (Ngwira et al., 2015) or “large” (Shao et al., 2024). Strengths are less than 0.05 V/km at 45 sites. 10% of the sites exceed 2.6 V/km, and the median strength is 0.51 V/km. Generally, Carrington-class geoelectric fields are strongest in the East and Midwest. A similar pattern, albeit with lower strengths, has been noted by Love et al. (2022) for the 13–14 March 1989 storm. The effects of uplifted igneous and metamorphic basement along the Rocky Mountain Front in Wyoming, Colorado, and New Mexico can also be seen in Figure 2. The geometric mean (e.g., Fleming & Wallace, 1986) of the site-by-site ratio of Carrington-class geoelectric-field strengths divided by those our model indicates were induced by the 13–14 March 1989 storm is 1.55. In other words, Carrington-class geoelectric-field strengths can be expected to be, on average, 55% greater than those of the 1989 storm.

In more detail, for our Carrington-class extrapolation, at survey site MEE62 (43.56°N, −70.21°E) in Maine, our median peak geoelectric-field strengths and corresponding 68% confidence interval are 34.69 [21.01, 57.26] V/km, at site VAQ58 in Virginia, our field strength and confidence interval are 30.30 [19.44, 47.20] V/km, but at another site just 204 km away, VAQ55 (37.88°N, −79.80°E) in Virginia, our field strength and interval are orders of magnitude lower, 0.05 [0.03, 0.07] V/km. This illustrates the high degree of geographic granularity exhibited by the induced geoelectric fields, a reflection of the geologic heterogeneity of the subsurface. VAQ58 is located atop resistive granite intrusions within the Piedmont physiographic province, whereas VAQ55 is located atop conductive shale within the Appalachian Valley and Ridge province. The highest-to-lowest geoelectric-field strength ratio is approximately 2900. We provide numerical values for Carrington-class peak geoelectric-field strengths for each survey site, together with model parameters and measures of statistical significance, in the Supporting Information S1; they can also be found at Love, Lucas, Kelbert, Rigler, et al. (2025).

6. Errors and Validation

In Figure 3, we show peak 1-min-resolution estimated peak geopotentials $V(D)$ for three example long lines and for the intensities D of all 40 storms in Table 1. We take the logarithmic transformation of the storm intensities and the geopotentials, $V(D) \rightarrow \ln V(\ln D)$, and, for each line, we fit a straight-line model $\ln \tilde{V}(\ln D)$ to these transformed values. For each line, the model provides a reasonably good fit to the peak

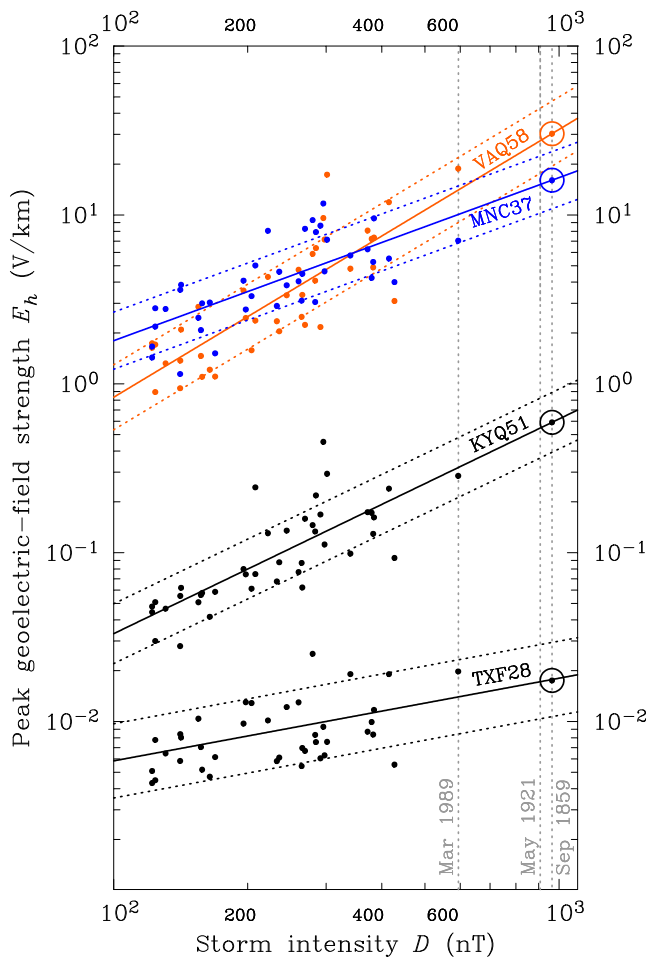


Figure 1. Estimated peak geoelectric-field strengths $E_h(D)$ at four survey sites: VAQ58 in Virginia (orange dots), MNC37 in Minnesota (blue dots), KYQ51 in Kentucky (black dots), and TXF28 in Texas (black dots), each for the intensities D of all 40 storms in Table 1. Model geoelectric-field strengths $\tilde{E}_h(D)$ for each survey site (solid lines) and corresponding 68% confidence envelope (dotted lines). Intensities of the March 1989, May 1921, and September 1859 storms (gray dotted lines), and extrapolated median peak geoelectric-field strengths for a Carrington-class storm at the four sites (dots with circles around them).

confidence interval is $[0.8297, 1.1755]$. In other words, our modeled geopotentials $\tilde{V}$, across many storms and many long lines, have an aggregate error of about 18%. Acknowledging the long chain of calculations and assumptions made here, we consider this to be a small error.

7. Comparisons With Industry Benchmarks

Whereas our map of 1-min-resolution peak geoelectric-field strength, Figure 2, is specifically for a storm with an intensity like that of the Carrington storm, benchmarks developed by the North American Electric Reliability Corporation (NERC) (2016) are for storms expected to occur, on average, once per century. Comparison is worthwhile. Consider, e.g., that the NERC benchmark field strength for Virginia latitudes in the Piedmont of the Appalachians is 2.34 V/km. This is an order of magnitude lower than the 1-min-resolution field strength of 30.30 V/km our model indicates for site VAQ58 in Virginia for a Carrington-class storm. Moreover, in our map, 10 of the 23 survey sites in Virginia, many in Eastern Virginia, have Carrington-class field strengths exceeding the NERC value.

estimated geopotentials, and the 68% confidence envelope, defined by the standard deviation of the residual differences between our straight line model and the logarithm of our estimated geopotentials, provides a good representation of dispersion; $\sigma = 0.3697$. From our models, we can, furthermore, obtain model peak geopotentials $\ln \tilde{V}(\ln D)$ corresponding to the 30 geopotential measurements listed in Table 2, and plotted on the map in Figure 4. For example, on the line between Mobridge, South Dakota, and Miles City, Montana, a geopotential of $V = 450$ V was measured during the May 1921 storm, but our model indicates that the median geopotential on that line for storms as intense as that of 1921 ($D = 907$ nT) is $\tilde{V} = 153$ V; the ratio $\tilde{V}/V = 0.3400$. For that line and that storm, the measured geopotential was considerably higher than our model predicts. The geometric mean, across all the lines and storms of the ratios $\tilde{V}/V$ is $\mu_g = 0.9876$. The arithmetic mean of the differences $\ln \tilde{V} - \ln V$ is $\mu_a = -0.0124 = \ln \mu_g$, and their standard deviation is $\sigma = 0.9540$, giving a large width factor of $\exp(\sigma) = 2.60$.

Several factors contribute to individual differences between modeled and measured geopotentials, including: (a) Storm-to-storm differences in local geomagnetic disturbance. This is likely considerable, but, in principle, the median geopotential of many storms should converge to our model. Additional data might resolve this. (b) The identification of peak geopotentials is likely crude. (c) Imprecise interpolation of geomagnetic field variation. A denser network of magnetometer stations would reduce these errors (Lucas et al., 2020). (d) Imprecise calculation of geoelectric fields obtained by combining geomagnetic field variation with magnetotelluric transfer functions; we believe that this source of error is relatively small (e.g., Kelbert et al., 2017; Love et al., 2019a). (e) Imprecise long-line voltages obtained by integrating geoelectric fields survey sites separated by ~ 70 km (Lucas et al., 2020). Denser surveying would reduce these errors (Love et al., 2022).

Although, we are not able to untangle these factors, we can examine their combined effect. For $n = 30$ differences, assumed to be normally distributed, the standard error (e.g., Altman & Bland, 2005) on the arithmetic mean is $\sigma/\sqrt{n} = 0.1772$. The 68% confidence interval on the arithmetic mean is $[\mu_a - \sigma/\sqrt{n}, \mu_a + \sigma/\sqrt{n}]$. With exponential transformation, we obtain the 68% confidence interval on the geometric mean: $[\mu_g/\exp(\sigma/\sqrt{n}), \mu_g \cdot \exp(\sigma/\sqrt{n})]$. Numerically, then, for $\mu_g = 0.9876$ the

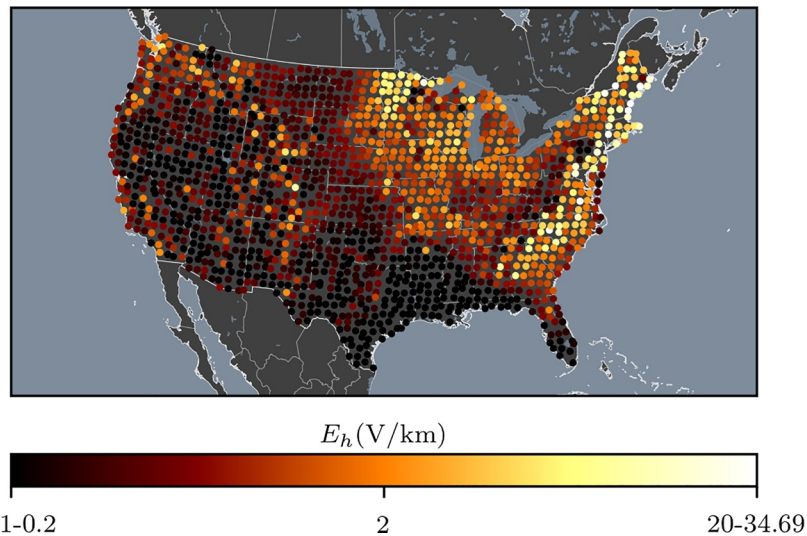


Figure 2. Map of median 1-min-resolution peak geoelectric-field strengths at each survey site across the United States as would be induced by an ensemble of magnetic storms as intense as the 2 September 1859 Carrington storm. Basemap by Met Office (2010–2015).

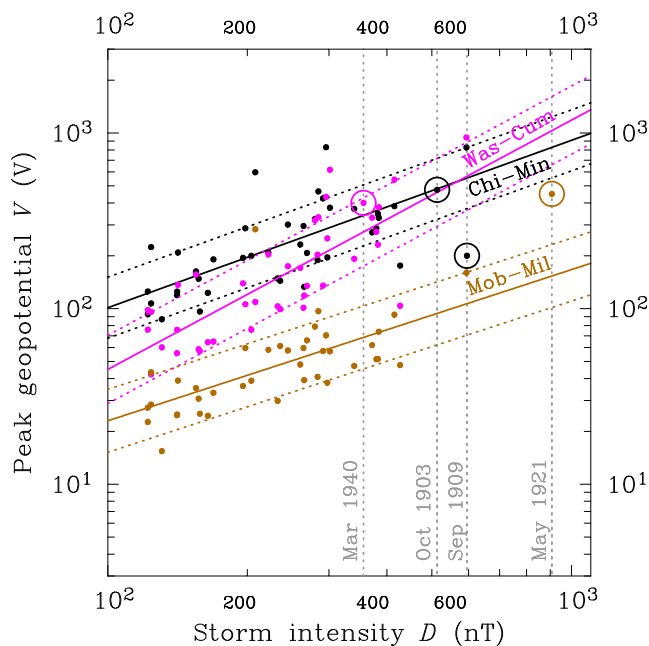


Figure 3. Estimated peak geopotentials $V(D)$ on three long lines: Washington to Cumberland (Was-Cum, pink dots), Chicago to Minneapolis (Chi-Min, black dots), Mobridge to Miles City (Mob-Mil, brown dots), each for the intensities D of all 40 storms in Table 1. Model geopotentials $\tilde{V}(D)$ for each line (solid lines) and corresponding 68% confidence envelope (dotted lines). Intensities of the October 1903, September 1909, May 1921, and March 1940 storms (gray dotted lines), and measured geopotentials on the color-coded lines (dots with circles around them).

A test advocated by NERC is estimation of GICs in transmission systems subjected to steady geoelectric fields. Consider the peak 1-min-resolution geoelectric field intensity we estimate was realized at site VAQ58 in Virginia: 30.30 V/km. On a 100 km line with resistance of 0.01 Ω /km, it would drive 3030 A of current for a minute. In comparison, Boteler (2001, Table 1) lists a peak GIC of 195 A for the Hydro-Québec system in March 1989. Although, in our map of Figure 2, other survey sites in Virginia exhibit lower geoelectric fields, this example illustrates the importance of making realistic treatments of the solid Earth—because of localized differences in surface impedance (subsurface geological structure), some locations would realize very extreme geoelectric fields. These could drive extreme currents should a power line be deployed in such a location. Time-dependent calculations using realistic impedances and geomagnetic monitoring time series are being pursued for the United States, although for storms much more modest than Carrington (e.g., Balch et al., 2023; Bernabeu, 2013).

8. Opportunities in Other Countries

Our results can be compared with those of Marshall et al. (2020) for Australia. They used a statistical method different from ours and assumed that a Carrington-class storm would have a much higher intensity. Despite this, we obtain geoelectric-field strengths for a Carrington-class storm that, at some locations (e.g., 34.69 V/km in Maine and 30.30 V/km in Virginia) are substantially higher than the 9.6 V/km that Marshall et al. obtained as a peak value across all of Australia. This might be further investigated. More generally, we imagine that maps, something like ours, might be constructed of Carrington-class geoelectric-field strengths for other countries where related research on storm-induced hazards is being performed (e.g., Cordell et al., 2025; Hübert et al., 2025; Juusola et al., 2025; Mac Manus et al., 2022; Malone-Leigh et al., 2024; Ribeiro et al., 2023; Shao et al., 2024; Torta

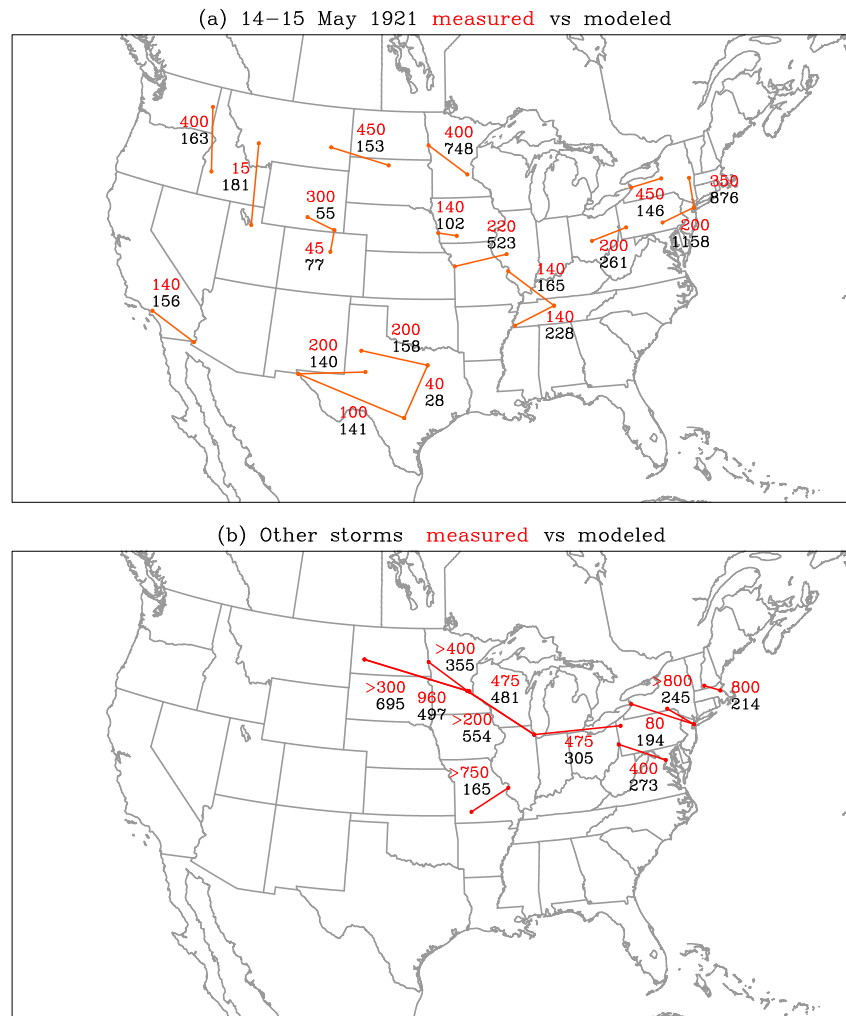


Figure 4. Maps showing long lines for (a) the May 1921 storm and (b) five other storms listed in Table 2: Peak measured geopotential values (V , red) and model values ($\tilde{V}$, black).

et al., 2023). Identification of independent data, such as geopotential measurements of the type used here, could help to validate the needed extrapolations.

9. Conclusions

When a Carrington-class storm occurs in the future, geoelectric fields across the United States are likely to be very strong in the East, in the vicinity of many of America's largest cities, and in the Midwest. Strengths are likely to exceed 5.00 V/km at many places; the highest strength we estimate is for a site in Maine, 34.69 [21.01,57.26] V/km. In Virginia, in the vicinity of Washington D.C., field strengths would likely range from as high as 30.30 [19.44,47.20] V/km to as low as 0.05 [0.03,0.07] V/km. This difference represents the high degree of geographic granularity exhibited by induced geoelectric fields, a result of complicated subsurface geological structure. Across the contiguous United States, a Carrington-class storm would likely induce geoelectric fields with strengths, on average, 55% greater than induced by the 13–14 March 1989 storm, a storm that caused significant interference to North American power-grid systems. Our model geoelectric-field strengths for a Carrington-class storm exceed, in many places, 100-year benchmark values used in testing the resilience of power-transmission systems.

Disclaimer

Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the United States Government.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Magnetometer data from observatories operated by NRCan and the USGS are available from the Kyoto (wdc.kugi.kyoto-u.ac.jp) and the Edinburgh (<http://www.wdc.bgs.ac.uk>) World Data Centers (Reay et al., 2013; Watanabe, 2009). Magnetotelluric transfer functions are available from the Incorporated Research Institutions for Seismology Data Management Center (ds.iris.edu/spud/emtf) (Kelbert et al., 2011, 2018). The Oulu *Dst* index is available from the University of Oulu, Finland (dcx.oulu.fi) (Karinen & Mursula, 2005; Mursula et al., 2008). Numerical values for Carrington-class peak geoelectric-field strengths for each survey site, together with model parameters and measures of statistical significance, in the Supporting Information S1; they can also be found at Love, Lucas, Kelbert, Rigler, et al. (2025).

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